

# Rarefied hypersonic cylinder

## Operator learning, baselines, and honest UQ

From author-supplied DSMC fields to a leakage-controlled teaching experiment

## OUTCOMES

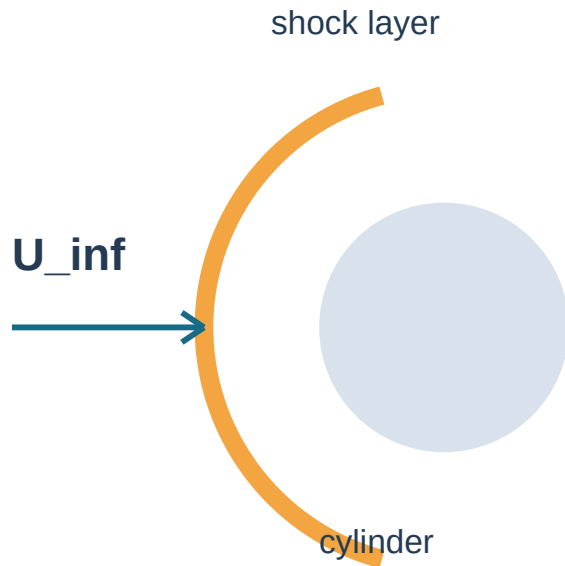
- Separate freestream Mach from the predicted local Mach field.
- Formulate a parameter-to-field operator for DSMC cylinder data.
- Freeze whole-case splits before model selection.
- Read the branch, trunk, layerwise fusion, and output head.
- Challenge the neural model with a structured interpolation baseline.
- Audit training error; a single MLP does not provide calibrated uncertainty.

## CLAIM

The lab executes a compact, author-released DSMC derivative. The default CPU model is a **teaching analog**; it does not reproduce the paper's full-resolution accuracy or speedup.

## RULE

**If a transparent baseline wins, retain that result. Complexity is not evidence.**



**Mach number:**  $M_{\text{inf}} = U_{\text{inf}}/a_{\text{inf}}$  controls compressibility. The target  $M(x,y)$  is a different, spatial field.

**Knudsen number:**  $Kn = \lambda/D$  measures rarefaction. As  $\lambda$  becomes non-negligible, continuum closure and no-slip assumptions need qualification.

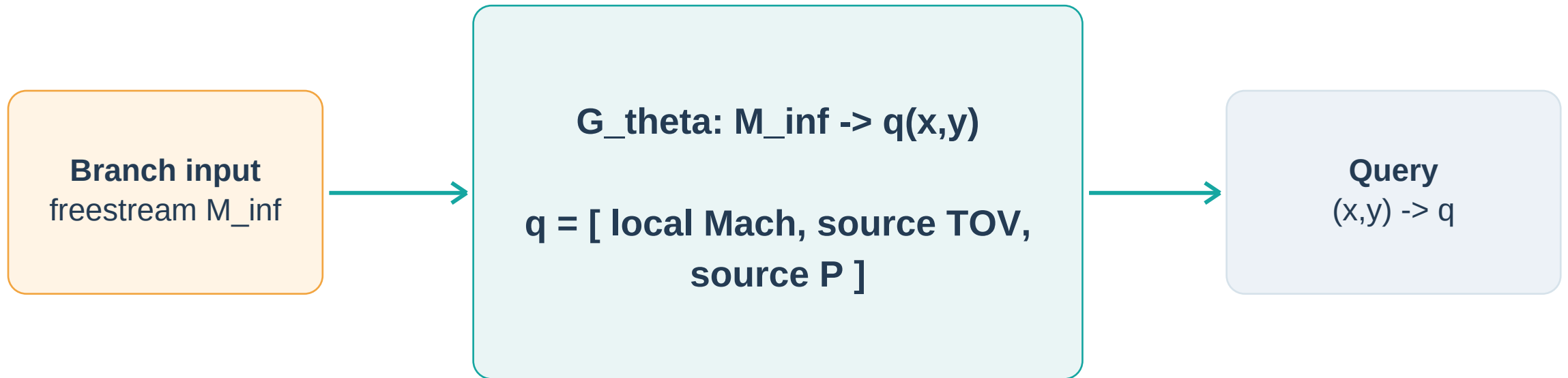
**Sampling scale:** DSMC fields estimate particle moments. A sharp shock and finite particle count make target noise spatially nonuniform.

**Physical motivation.** At high Mach, appropriately nondimensionalized fields can vary slowly with Mach under compatible gas, rarefaction, geometry and boundary conditions. This motivates interpolation; a small error alone is not proof of Mach independence. The source TOV/P scaling must be verified.

**Required model.** Three 96-unit tanh layers; inputs (Mach,  $x$ ,  $y$ ); equal standardized-target MSE. Scalers use training cases only. Adam, fixed seed 760, maximum 300 epochs and 40-epoch patience; best epoch selected on four whole validation cases. Report training error as an underfitting diagnostic.

**Provenance.** Research DSMC data precede the course. This MLP is a new teaching model, not the paper's trained Fusion-DeepONet. A single fit provides no calibrated UQ. The committed dataset is CC BY 4.0, authorized by Ehsan Roohi on 2026-09-05; see DATA\_LICENSE.md.

# The learning object is an operator



**A field surrogate must be judged at unseen flow cases, not merely unseen grid points from a case already used during training.**

# Data audit: small derivative, complete provenance

**20**

Mach cases

**400 x 400**

source grid per case

**50 x 50**

deterministic selection

**44,500**

finite retained points

**0.5 MB**

committed NPZ

**KEPT**

- M\_inf, x, y
- local Mach, source TOV, source P (legacy ratio keys do not verify scaling)
- case ID and original source-grid row
- archive and derivative SHA-256 hashes

**EXCLUDED**

- 1.4 GB source ZIP and 4.9 GB expanded files
- training logs and duplicate checkpoints
- historical scripts with inconsistent protocols
- research checkpoints not needed for the lab

# Whole-case splitting prevents a false claim

**TRAIN**

5, 6, 7, 8, 9, 10, 11, 12,  
13, 14

**VALIDATION**

8.25, 8.75, 9.25, 9.75

**BLIND INTERP.**

5.5, 6.5, 7.5, 8.5, 9.5

**BLIND EXTRA.**

15

**Wrong question:** can the network fill random points from a Mach case it has already seen?

**Scientific question:** can it predict a complete field at an operating condition excluded from fitting and tuning?

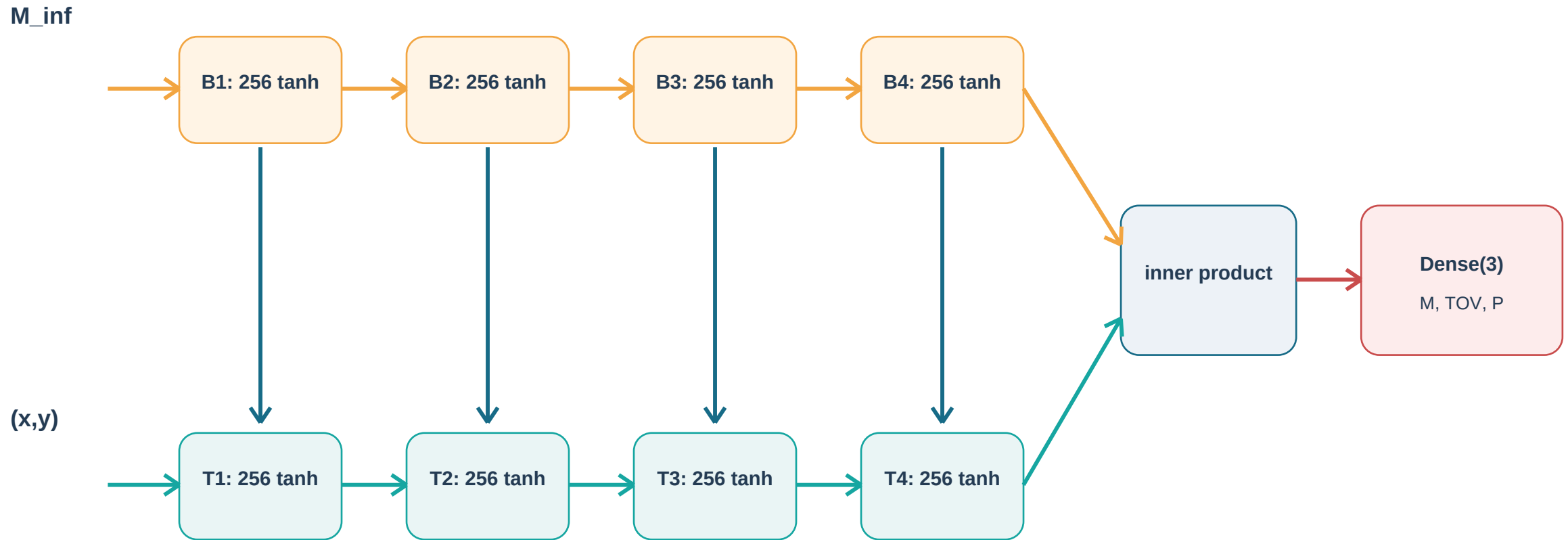
# The mandatory baseline is field interpolation

$$\hat{q}(M_q) = q(M_a) + [(M_q - M_a)/(M_b - M_a)] [q(M_b) - q(M_a)]$$

| Split                 | Local Mach | Temperature | Pressure |
|-----------------------|------------|-------------|----------|
| Blind interpolation   | 0.398%     | 0.609%      | 0.779%   |
| Mach 15 extrapolation | 0.538%     | 0.877%      | 0.848%   |

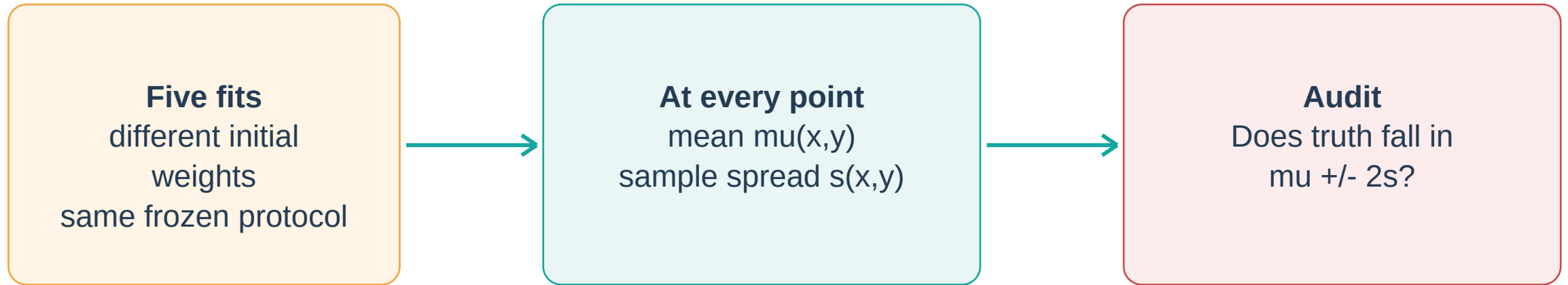
**Result: a neural operator must beat a sub-1% structured baseline here, or justify itself through a different deployment constraint.**

# Reviewed Fusion-DeepONet topology



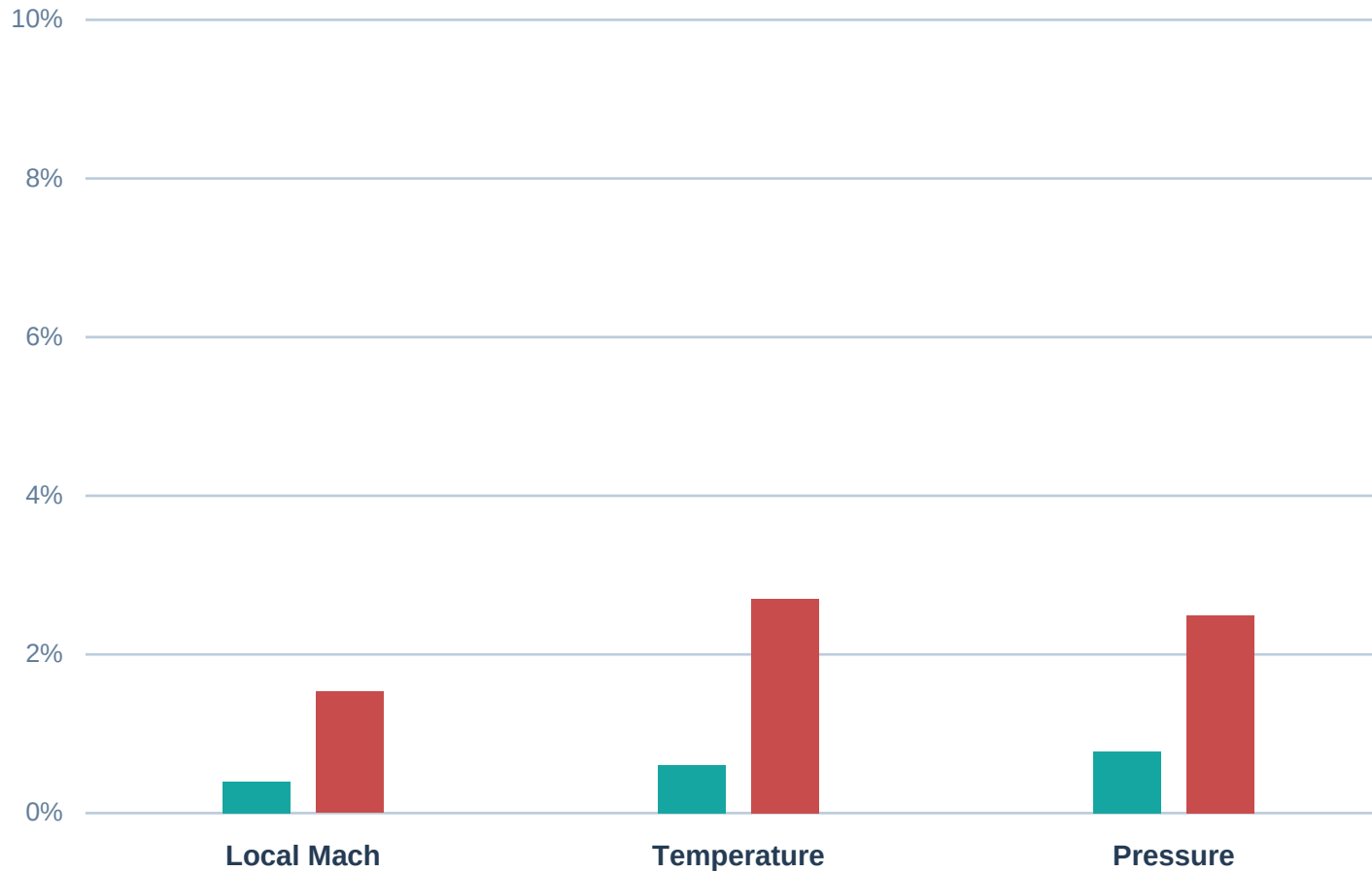
Each block includes dropout in the reviewed stored model. Branch state is added to the matching trunk depth before the final branch-trunk combination.

- 1 Fit scalers** Training cases only; MinMax inputs and standardized targets.
- 2 Optimize** Adam with validation-monitored early stopping and learning-rate reduction.
- 3 Weight** Reviewed archive loss:  $\text{MSE}_M + \text{MSE}_T + 5 \text{ MSE}_p$  in standardized space.
- 4 Repeat** Five independent initializations; preserve member predictions.
- 5 Open blind cases** After design is frozen; report interpolation and extrapolation separately.



**Low coverage means overconfidence. High coverage can also be uninformative if intervals are too wide. Report coverage and interval width, by target and by split.**

# Executed evidence: trained MLP versus interpolation



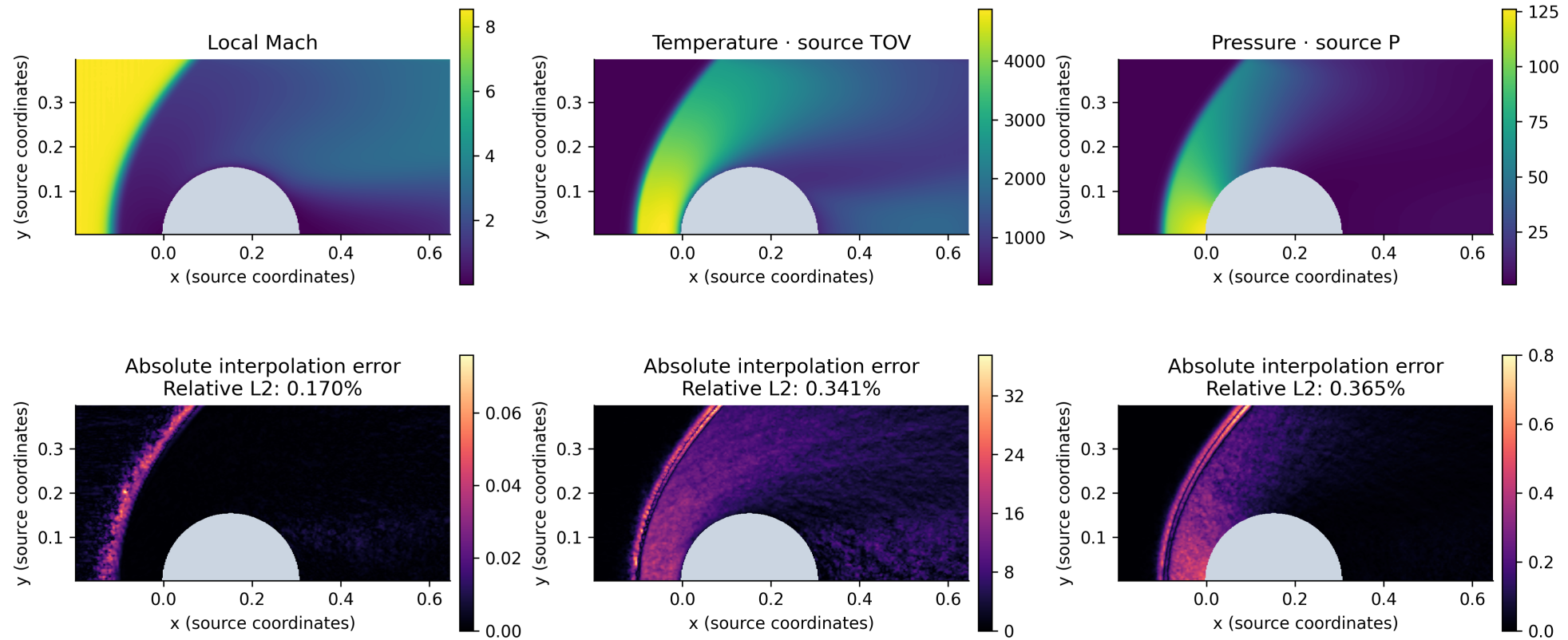
## Baseline

0.398% | 0.609% | 0.779%

## Trained 3x96 MLP

1.53% | 2.70% | 2.50%

## Rarefied cylinder | freestream Mach 8.5 Original 400 × 400 DSMC fields



# Decision gate: when is a neural operator justified?

## Accuracy

Does it beat field interpolation on frozen blind cases?

## Geometry

Must one model span grids, shapes, or boundary conditions?

## Availability

Will deployment lack bracketing high-fidelity solutions?

## Cost

Do many field queries amortize training and data production?

## Physics

Are shock location, surface loads, positivity, and limits acceptable?

## Uncertainty

Is spread calibrated on the deployment regime?

**SUBMIT**

- Baseline and teaching-operator relative-L2 errors by target and split.
- One physics-based interpretation of an error hot spot.
- Training error and an explicit statement that the single MLP has no calibrated UQ.
- A short explanation of random-point leakage.
- One deployment condition that could justify full neural training.

**MAY CLAIM**

The compact DSMC derivative, frozen split, baseline, and CPU teaching analog were executed.

**MAY NOT CLAIM**

Published full-model accuracy, speedup, complete-field reproduction, or calibrated UQ.

- Roohi et al. (2026), Neural Networks for Rarefied Gas Dynamics: Relaxation Problem, Polyatomic Shock Waves, and Hypersonic Cylinder Flow. Physics of Fluids 38, 057108. <https://doi.org/10.1063/5.0334590>
- Lu et al. (2021), Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. Nature Machine Intelligence 3, 218-229. <https://doi.org/10.1038/s42256-021-00302-5>
- Lakshminarayanan, Pritzel, and Blundell (2017), Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 30.
- Bird (1994), Molecular Gas Dynamics and the Direct Simulation of Gas Flows. Oxford University Press.

Rebuild data: `qa/build_hypersonic_cylinder_subset.py` | Rebuild evidence: `qa/run_hypersonic_cylinder_evidence.py` | Source archive SHA-256 begins `bda221759a37`